An explicit dimension-free isoperimetric plateau in the cube

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Abstract

We prove that, in every dimension, the relative Euclidean isoperimetric profile of the unit cube equals one whenever the prescribed volume differs from one half by at most 6.25×10^{-12}. The only minimizers in this interval are coordinate slabs. This answers Glaudo's question on the existence of a dimension-free central plateau. The proof combines a dimension-free local optimality estimate for slabs, established through BV slicing and a small-support estimate, with a quantitative Gaussian rigidity argument that forces the normal of an approximating half-space to be close to a coordinate direction.

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1 Introduction

For the open unit cube $Q_n = (0,1)^n$, endowed with Lebesgue probability measure, define

$$P_n(E) = \mathrm{Per}(E; Q_n), \qquad I_n(a) = \inf\{P_n(E) : E \subset Q_n,\ |E| = a\},$$

where $|E|$ is the volume of E and the perimeter $\mathrm{Per}(E; Q_n)$ is relative to the open cube: portions of the boundary on its faces are not counted. Define $I_\infty = \inf_{n\geq 1} I_n = \lim_{n\to\infty} I_n$. Sets are considered up to null sets. Coordinate slabs of any volume $a \in (0,1)$ have perimeter one, so $I_n(a) \leq 1$. The question is whether they minimize on a common interval about half-volume as n varies.

Glaudo [7] proves a central plateau in each fixed dimension and credits it to the earlier variational work of Acerbi, Fusco, and Morini [1]. This result was part of an open problem stated by Brezis and Bruckstein [5, Open Problem 10.1] (see also [6, Remark 2]). Glaudo [7]

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explicitly formulates the dimension-uniform version of this question as Open Question 1.1 in [7]. He also proves that the limiting cube profile stays strictly above the elementary Gaussian lower bound away from half-volume.

We establish the dimension-uniform central statement with the explicit constant.

Theorem 1 (Central plateau with explicit width)**.**

$$I_n(a) = 1 \quad \textit{for} \quad |a - 1/2| \le \varepsilon_0, \qquad \varepsilon_0 = \frac{1}{160\,000\,000\,000} = 6.25 \times 10^{-12}. \tag{1}$$

The only minimizers are slabs attached to cube faces.

The precise statement version of this theorem is Theorem 8. This answers the question of Glaudo.

The proof combines two estimates, both uniform in the dimension. The first establishes local optimality of coordinate slabs. If $S \subset Q_n$ is a coordinate slab of volume $a \in [0.49, 0.51]$, then

$$P_n(E) \ge P_n(S) = 1$$

for every set $E \subset Q_n$ of finite relative perimeter satisfying

$$|E| = a, \qquad |E \triangle S| \le 10^{-4}.$$

The argument combines BV slicing with a small-support estimate to control the loss of perimeter caused by empty and full fibers. The second estimate is a quantitative rigidity statement. If

$$P_n(E) \le 1, \qquad \big||E| - 1/2\big| \le 10^{-3},$$

then there exists a coordinate slab S with $|S| = |E|$ such that

$$|E \triangle S| \le 40\sqrt{\big||E| - 1/2\big|}.$$

To prove this, we pull E back to Gaussian space under the coordinatewise Gaussian distribution map and apply the Gaussian barycenter stability theorem of Barchiesi, Brancolini, and Julin [3]. Based on these two estimates, for $\big||E| - 1/2\big| \le 6.25 \times 10^{-12}$, we show that the rigidity estimate places every competitor of perimeter at most one within the neighborhood covered by the local optimality estimate. Applying that estimate gives $P_n(E) \ge 1$.

2 Preliminaries and Gaussian normalization

We use BV slicing, coarea, and chain rules for relative variation; see [2, 8]. Expectations on cubes are Lebesgue integrals, and let

$$\varphi(t) = \frac{e^{-t^2/2}}{\sqrt{2\pi}}, \quad \Phi(t) = \int_{-\infty}^{t} \varphi(s)\, ds, \quad \kappa = \sqrt{2\pi}, \quad I_{\mathrm{G}}(a) = \varphi(\Phi^{-1}(a)).$$

Extend I_{G} by zero at the endpoints and set $J = \kappa I_{\mathrm{G}}$. Let γ_n be standard Gaussian probability measure with density φ_n. Our Gaussian perimeter normalization is

$$p_\gamma(F) = \int_{\partial^* F} \varphi_n(x)\, d\mathcal{H}^{n-1}(x),$$

where $\partial^* F$ is the reduced boundary of F. The half-space $H_{\omega,z} = \{x : \omega \cdot x < z\}$, $|\omega| = 1$, has measure $\Phi(z)$ and perimeter $\varphi(z)$.

The map $T : \mathbb{R}^n \to Q_n$ defined by $T(x) = (\Phi(x_1), \dots, \Phi(x_n))$ is a smooth diffeomorphism. It pushes γ_n onto Lebesgue probability measure on Q_n: for every bounded Borel function b,

$$\int_{\mathbb{R}^n} b(T(x))\, d\gamma_n(x) = \int_{Q_n} b(y)\, dy.$$

This follows by the one-dimensional substitution $y_i = \Phi(x_i)$, $dy_i = \varphi(x_i)\, dx_i$, in every coordinate.

If E has finite relative perimeter and $F = T^{-1}(E)$, then F has locally finite perimeter. Indeed, T and its inverse have bounded derivatives on corresponding compact subsets, so local changes of variables preserve the BV property. The finite-perimeter area formula gives

$$P_n(E) = \int_{\partial^* F} |\operatorname{cof} DT(x)\, \nu_F(x)|\, d\mathcal{H}^{n-1}(x),$$

where ν_F is the measure-theoretic outward unit normal, $D\mathbf{1}_F = -\nu_F\, \mathcal{H}^{n-1} \lfloor_{\partial^* F}$, and

$$\operatorname{cof} DT = (\det DT)(DT)^{-\top}.$$

Here $A^{-\top} := (A^{-1})^\top$, and the cofactor matrix acts on a normal to give the oriented surface-area scaling. Since

$$DT = \operatorname{diag}(\varphi(x_1), \dots, \varphi(x_n)), \qquad \det DT = \varphi_n(x),$$

its ith normal component is $\varphi_n(x)\nu_i/\varphi(x_i)$. Consequently,

$$|\operatorname{cof} DT(x)\nu| = \varphi_n(x) \sqrt{\sum_i \frac{\nu_i^2}{\varphi(x_i)^2}} = \kappa \varphi_n(x) \sqrt{\sum_i \nu_i^2 e^{x_i^2}}.$$

Thus, with

$$W(x, \nu) = \sqrt{\sum_i \nu_i^2 e^{x_i^2}}, \tag{2}$$

$$P_n(E) = \kappa \int_{\partial^* F} W(x, \nu_F)\, d\sigma_F, \qquad d\sigma_F = \varphi_n\, d\mathcal{H}^{n-1}. \tag{3}$$

For completeness, the global formula uses the local identity

$$\operatorname{Per}(E; T(B_R)) = \int_{\partial^* F \cap B_R} |\operatorname{cof} DT\, \nu_F|\, d\mathcal{H}^{n-1}$$

and continuity from below as $R \to \infty$. This restricts perimeter measures to open regions and adds no artificial boundary on ∂B_R.

Since $|\nu_F| = 1$ and $e^{x_i^2} \geq 1$, $W \geq 1$. In particular, $p_\gamma(F) \leq P_n(E)/\kappa < \infty$. Gaussian isoperimetry and $\gamma_n(F) = |E|$ give

$$P_n(A) \geq \kappa p_\gamma(T^{-1}(A)) \geq \kappa I_{\mathrm{G}}(|A|) = J(|A|). \tag{4}$$

The assertion is automatic for sets of infinite perimeter.

We shall also use the functional form of this comparison. Bobkov's Gaussian inequality [4, Corollary 1] reads

$$I_{\mathrm{G}}(\mathbb{E}_{\gamma_d} g) \leq \mathbb{E}_{\gamma_d} \sqrt{I_{\mathrm{G}}(g)^2 + |\nabla g|^2}, \qquad g : \mathbb{R}^d \to [0,1] \text{ Lipschitz}.$$

For $g = f \circ T$, the chain rule holds almost everywhere and gives

$$|\nabla (f \circ T)(x)|^2 = \sum_i \varphi(x_i)^2 |\partial_i f(T(x))|^2 \leq \kappa^{-2} |\nabla f(T(x))|^2.$$

Multiplying Bobkov's inequality by κ and changing variables therefore yields

$$J(\mathbb{E} f) \leq \mathbb{E} \sqrt{J(f)^2 + |\nabla f|^2}, \qquad f : Q_d \to [0,1] \text{ Lipschitz}. \tag{5}$$

All expectations in this formula are uniform expectations on Q_d. One can also recover (4) from (5). Choose $[0,1]$-valued Lipschitz functions f_j converging strictly in BV to $\mathbf{1}_A$, so that $f_j \to \mathbf{1}_A$ in L^1 and $\int |\nabla f_j| \to P_n(A)$; such approximation is available on the cube [2]. Since J is bounded, continuous, and zero at both endpoints, $\int J(f_j) \to 0$. Using $\sqrt{J(f_j)^2 + |\nabla f_j|^2} \leq J(f_j) + |\nabla f_j|$ and passing to the limit gives the perimeter bound.

Differentiating the Gaussian profile, with $z = \Phi^{-1}(a)$, gives

$$I_{\mathrm{G}}'(a) = -z, \qquad I_{\mathrm{G}}''(a) = -\frac{1}{I_{\mathrm{G}}(a)}, \qquad J''(a) = -\frac{\kappa^2}{J(a)} = -\frac{2\pi}{J(a)}.$$

Thus J is concave, symmetric about $1/2$, satisfies $J(1/2) = 1$, and takes values in $[0,1]$.

For a set F with $\gamma_n(F) = \Phi(z)$, write

$$B(F) = \int_F x \, d\gamma_n, \qquad D = p_\gamma(F) - \varphi(z), \qquad \beta = \varphi(z) - |B(F)|. \tag{6}$$

To see $\beta \geq 0$, take $\omega = -B(F)/|B(F)|$ when $B(F) \neq 0$, and put $H = H_{\omega,z}$. The one-dimensional Gaussian moment satisfies $\int_{-\infty}^{z} s\varphi(s)\, ds = -\varphi(z)$, hence $B(H) = -\varphi(z)\omega$. Equal volumes give

$$\begin{aligned} \varphi(z) - |B(F)| &= \int_{\mathbb{R}^n} (\mathbf{1}_F - \mathbf{1}_H)(\omega \cdot x - z)\, d\gamma_n \\ &= \int_{F \triangle H} |\omega \cdot x - z|\, d\gamma_n \geq 0. \end{aligned}$$

The integrands have the asserted sign because $F \setminus H$ lies above the plane and $H \setminus F$ lies below it. If $B(F) = 0$, nonnegativity is immediate.

The Main Theorem of [3] implies

$$\beta \le 160\pi^3(1+z^2)D. \tag{7}$$

Here the published perimeter is κp_γ, its deficit is κD, and its numerical constant is $80\pi^2\kappa$. The barycenter is unchanged. Thus its bound becomes $80\pi^2\kappa(1+z^2)\kappa D = 160\pi^3(1+z^2)D$. For $B(F) \neq 0$, Proposition 4 of the same reference applies to the half-space aligned with $-B(F)$, not just a half-space minimizing the symmetric difference, and yields

$$\eta^2 \le 4e^{-z^2/2}\beta, \qquad \eta = \gamma_n(F \triangle H_{\omega,z}). \tag{8}$$

These are the external quantitative Gaussian estimates used below.

We record also the Gaussian Gauss–Green formula used in the proof. For a bounded C^1 vector field X such that $\operatorname{div}_\gamma X := \operatorname{div} X - x \cdot X$ is Gaussian integrable,

$$\int_F \operatorname{div}_\gamma X \, d\gamma_n = \int_{\partial^* F} X \cdot \nu_F \, d\sigma_F. \tag{9}$$

For compactly supported X, this follows from Euclidean Gauss–Green applied to $\varphi_n X$, since $\operatorname{div}(\varphi_n X) = \varphi_n \operatorname{div}_\gamma X$. For general X as above, insert smooth cutoffs χ_R with $0 \le \chi_R \le 1$, $\chi_R \to 1$, and $|\nabla \chi_R| \le C/R$. The extra term $\int_F \nabla\chi_R \cdot X \, d\gamma_n$ tends to zero; the remaining volume integral converges by dominated convergence and the boundary integral by $p_\gamma(F) < \infty$. In particular, applying this to constant vectors gives

$$\int_{\partial^* F} \nu_F \, d\sigma_F = -B(F). \tag{10}$$

3 BV slicing and local optimality of slabs

Let $BV(Q_d; [0,1])$ denote the set of $[0,1]$-valued functions on Q_d having bounded variation. For $f \in BV(Q_d; [0,1])$, the Lebesgue decomposition of its distributional derivative gives

$$Df = \nabla f \, \mathcal{L}^d + D^s f.$$

Here, Df is the distributional derivative of f, i.e., an $\mathbb{R}^d$-valued finite Radon measure defined by

$$\int_{Q_d} f \, \operatorname{div} \phi \, dx = -\int_{Q_d} \phi \cdot \, dDf, \qquad \phi \in C_c^1(Q_d; \mathbb{R}^d);$$

$\mathcal{L}^d$ is d-dimensional Lebesgue measure; $\nabla f = \dfrac{d(Df)^a}{d\mathcal{L}^d} \in L^1(Q_d; \mathbb{R}^d)$ is the density of the absolutely continuous part which agrees almost everywhere with the approximate gradient

of f; $D^s f$ is the singular part, satisfying $D^s f \perp \mathcal{L}^d$, which is concentrated on a set of Lebesgue measure zero. The total variation of Df is

$$|Df|(Q_d) = \int_{Q_d} |\nabla f(y)|\, dy + |D^s f|(Q_d).$$

Put $\psi(t) = \sqrt{1+t^2} - 1$. For $f \in BV(Q_d; [0,1])$ set

$$G(f) = \int_{Q_d} \psi(|\nabla f|)\, dy + |D^s f|(Q_d), \qquad s(f) = |\{f = 0\}| + |\{f = 1\}|. \tag{11}$$

Lemma 2 (Fiber-volume bound)**.** *For a finite-perimeter $E \subset (0,1) \times Q_d$, let*

$$E_y := \{x \in (0,1) : (x,y) \in E\}$$

be its one-dimensional fiber and $f(y) = |E_y|$ be the length of E_y. Then $f \in BV(Q_d; [0,1])$ and

$$P_{d+1}(E) \geq 1 - s(f) + G(f). \tag{12}$$

Proof. Write $\Omega = (0,1) \times Q_d$ and use coordinates (x,y). Throughout this proof boundary integrals are restricted to $\partial^* E \cap \Omega$, and $\nu_E = (\nu_x, \nu_y)$. Let $\pi(x,y) = y$ and $p(y) = \mathrm{Per}(E_y; (0,1))$ for almost every y. The BV slicing theorem [2, Section 3.11] gives the measure identity

$$\pi_{\#}\big(|\nu_x| \mathcal{H}^d \lfloor_{\partial^* E \cap \Omega}\big) = p(y)\, dy. \tag{13}$$

In particular p is measurable, integrable, and $\int p\, dy \leq P_{d+1}(E)$.

By the Gauss–Green formula for sets of finite perimeter and Fubini's theorem (see [2, Chapter 3]), the transverse signed pushforward satisfies

$$\pi_{\#}\big(\nu_y \mathcal{H}^d \lfloor_{\partial^* E \cap \Omega}\big) = -Df. \tag{14}$$

Indeed, for $\zeta \in C_c^1(Q_d; \mathbb{R}^d)$, test the distributional derivative of $\mathbf{1}_E$ against $(0, \eta_j(x)\zeta(y))$, where $\eta_j \in C_c^1(0,1)$, $0 \leq \eta_j \leq 1$, and $\eta_j \to 1$ pointwise. There is no derivative of η_j in the divergence of this transverse field. The Gauss–Green formula and dominated convergence therefore give

$$\int_{Q_d} f\ \mathrm{div}\, \zeta\, dy = \int_{\partial^* E \cap \Omega} \zeta(y) \cdot \nu_y\, d\mathcal{H}^d.$$

The right side is bounded by $P_{d+1}(E)\|\zeta\|_\infty$. Since $0 \leq f \leq 1$ and f is integrable, this proves $f \in BV(Q_d)$. By the definition of Df, the left side above is also equal to $-\int_{Q_d} \zeta(y) \cdot dDf(y)$, yielding the formula in (14).

Combine the two measures into $\mu = (p\, dy, -Df)$, an $\mathbb{R}^{d+1}$-valued measure on Q_d. For bounded Borel $\alpha \geq 0$ and $\zeta : Q_d \to \mathbb{R}^d$ satisfying $\alpha^2 + |\zeta|^2 \leq 1$, Cauchy–Schwarz gives

$$\alpha|\nu_x| + \zeta \cdot \nu_y \leq \sqrt{\alpha^2 + |\zeta|^2}\sqrt{|\nu_x|^2 + |\nu_y|^2} \leq 1.$$

The pushforward identities in (13) and (14) imply

$$\int \alpha p \, dy - \int \zeta \cdot dDf = \int_{\partial^* E \cap \Omega} (\alpha(y)|\nu_x| + \zeta(y) \cdot \nu_y) \, d\mathcal{H}^d \leq P_{d+1}(E).$$

The supremum over these fields equals $|\mu|(Q_d)$ by the total-variation duality for vector measures. Requiring $\alpha \geq 0$ loses nothing: replacing α by $|\alpha|$ preserves the norm constraint and increases the pairing because $p \, dy \geq 0$. The Lebesgue decomposition $\mu = (p, -\nabla f) \, dy + (0, -D^s f)$ consists of mutually singular parts, so

$$P_{d+1}(E) \geq |\mu|(Q_d) = \int_{Q_d} \sqrt{p^2 + |\nabla f|^2} \, dy + |D^s f|(Q_d). \tag{15}$$

This is the multidimensional-parameter form of the standard fiber-perimeter estimate; compare [9, Proposition 2.5], where the parameter is one-dimensional. The argument above supplies the version needed here for arbitrary d.

For almost every y, the one-dimensional set E_y has finite perimeter. Its characteristic function has finitely many jumps inside $(0, 1)$, each of size one. If $0 < f(y) < 1$, it cannot be constant, and at least one interior jump is present. Thus $p(y) \geq \mathbf{1}_{\{0<f(y)<1\}}$. If $f(y) = 0$ or 1, E_y is empty or full up to a null set and $p(y) = 0$. BV locality says that the approximate gradient of f vanishes almost everywhere on each of the level sets $\{f = 0\}$ and $\{f = 1\}$. Consequently, with $A = \{0 < f < 1\}$,

$$\begin{aligned} P_{d+1}(E) &\geq \int_A \sqrt{1 + |\nabla f|^2} \, dy + |D^s f|(Q_d) \\ &= |A| + \int_{Q_d} (\sqrt{1 + |\nabla f|^2} - 1) \, dy + |D^s f|(Q_d) \\ &= 1 - s(f) + G(f). \end{aligned}$$

The singular contribution $|D^s f|$ is kept explicitly in the variation estimate. The locality argument is used only to identify the absolutely continuous density ∇f on the relevant regions. □

For $0 < q \leq 1/2$, let

$$H(q) = J(q)/q.$$

Concavity of J implies $J(t) \geq H(q)t$ for $0 \leq t \leq q$. For any nonnegative BV function v with $|\{v > 0\}| \leq q$, let $a_t = |\{v > t\}| \leq q$. By (4) and the coarea formula [2, Theorem 3.40],

$$\begin{aligned} |Dv|(Q_d) &= \int_0^\infty \mathrm{Per}(\{v > t\}; Q_d) \, dt \geq \int_0^\infty J(a_t) \, dt \\ &\geq H(q) \int_0^\infty a_t \, dt = H(q) \int_{Q_d} v \, dy. \end{aligned} \tag{16}$$

Moreover $H(q) \to \infty$ as $q \downarrow 0$. Indeed, writing $q = \overline{\Phi}(t)$ gives $t \to \infty$ and

$$\overline{\Phi}(t) = \int_t^\infty \varphi(s)\, ds \le \frac{1}{t} \int_t^\infty s\varphi(s)\, ds = \frac{\varphi(t)}{t}.$$

Hence $H(q) = \kappa\varphi(t)/\overline{\Phi}(t) \ge \kappa t \to \infty$. The coarea formula above uses the full measure $|Dv|$, including its singular part. No estimate of the form $|Dv| \ge J(\int v)$ for general nonnegative v is needed.

Lemma 3 (Saturation compensation)**.** *Let $a = \mathbb{E}f$, $0 < h < \min(a, 1-a)$, and $0 < q \le 1/2$. If*

$$\int |f - a| \le (\min(a, 1-a) - h)q,$$

then, provided $C(h,q) := \frac{\sqrt{3}hH(q)}{4} - \frac{1}{2} \ge 0$,

$$G(f) \ge C(h,q)s(f), \tag{17}$$

where G, s were defined in (11).

Proof. Set $b = \min(a, 1-a) - h > 0$ and

$$g_- = (h - f)_+, \qquad g_+ = (f - 1 + h)_+, \qquad \lambda = \sqrt{3}/2.$$

Since $h < 1/2$, the positivity sets $A_- = \{f < h\}$ and $A_+ = \{f > 1-h\}$ are disjoint. On A_-, $|f - a| > a - h \ge b$, and on A_+, $|f - a| > 1 - a - h \ge b$. Therefore, by Markov inequality,

$$|A_\pm| \le |\{|f - a| > b\}| \le b^{-1} \int |f - a| \le q.$$

Here and below the small-support condition concerns these positivity sets, rather than topological supports.

Allocation of the singular variation. Define the C^1 functions

$$\Psi_-(u) = \frac{(h-u)_+^2}{2h}, \qquad \Psi_+(u) = \frac{(u-1+h)_+^2}{2h}, \quad 0 \le u \le 1.$$

Their derivatives satisfy

$$\Psi'_-(u) = -\frac{(h-u)_+}{h}, \qquad \Psi'_+(u) = \frac{(u-1+h)_+}{h}, \qquad |\Psi'_-(u)| + |\Psi'_+(u)| \le 1. \tag{18}$$

For $0 \le u \le v \le 1$, the fundamental theorem of calculus yields

$$\sum_\pm |\Psi_\pm(v) - \Psi_\pm(u)| \le \int_u^v \sum_\pm |\Psi'_\pm(s)|\, ds \le v - u. \tag{19}$$

Write the singular BV derivative as $D^s f = D^j f + D^c f$, where $D^j f$ is the jump part and $D^c f$ is the Cantor part [2, Section 3.9]. The BV chain rule [2, Theorem 3.96] gives

$$D^c(\Psi_\pm \circ f) = \Psi'_\pm(\widetilde{f}) D^c f, \qquad D^j(\Psi_\pm \circ f) = (\Psi_\pm(f^+) - \Psi_\pm(f^-))\nu_f \mathcal{H}^{d-1} \restriction_{J_f} .$$

Here $\widetilde{f}$ is the approximate value outside the jump set, $f^\pm$ are the traces, ν_f is the chosen jump normal, and J_f is the approximate jump set. The derivative bound in (18) controls the sum of the Cantor variations, and the preceding two-point inequality in (19) controls the sum of the jump variations. The jump and Cantor measures are mutually singular, implying

$$|D^s(\Psi_\pm \circ f)| = |D^c(\Psi_\pm \circ f)| + |D^j(\Psi_\pm \circ f)|.$$

It follows that

$$\begin{aligned}
\sum_\pm |D^s(\Psi_\pm \circ f)|(Q_d) &= \sum_\pm |D^c(\Psi_\pm \circ f)|(Q_d) + |D^j(\Psi_\pm \circ f)|(Q_d) \\
&= \sum_\pm |\Psi'_\pm(\widetilde{f}) D^c f|(Q_d) + |(\Psi_\pm(f^+) - \Psi_\pm(f^-))\nu_f \mathcal{H}^{d-1} \restriction_{J_f}|(Q_d) \\
&\le |D^c f|(Q_d) + |f^+ - f^-| \, \mathcal{H}^{d-1} \restriction_{J_f} (Q_d) \\
&= |D^c f|(Q_d) + |D^j f|(Q_d) = |D^s f|(Q_d).
\end{aligned}$$

Since $\Psi_\pm \circ f = g_\pm^2/(2h)$, this reads

$$\sum_\pm |D^s(g_\pm^2)|/(2h) \le |D^s f|. \tag{20}$$

The absolutely continuous energy. For $\psi(t) = \sqrt{1+t^2} - 1$, $t \ge 0$, the convex conjugate is $\psi^*(v) = 1 - \sqrt{1-v^2}$ for $0 \le v \le 1$. For $v < 1$, this follows by maximizing $vt - \psi(t)$ at $t = v/\sqrt{1-v^2}$; the endpoint follows by a limit. Thus Fenchel's inequality is $\psi(t) \ge vt - \psi^*(v)$. Recall $\lambda = \frac{\sqrt{3}}{2}$. For $0 \le r \le 1$,

$$\psi^*(\lambda r) = \frac{\lambda^2 r^2}{1 + \sqrt{1 - \lambda^2 r^2}} \le \frac{(3/4) r^2}{1 + 1/2} = \frac{r^2}{2}.$$

On $A_\pm$ set $r = g_\pm/h$, $v = \lambda r = \lambda g_\pm/h$, and $t = |\nabla f|$. The chain rule for the absolutely continuous density gives $|\nabla(g_\pm^2)| = 2 g_\pm |\nabla f|$ there, and the gradient vanishes outside the corresponding active range almost everywhere. Summing over the disjoint sets $A_\pm$, and discarding the nonnegative energy on their complement, gives

$$\int \psi(|\nabla f|) \ge \sum_\pm \left\{ \frac{\lambda}{2h} \int |\nabla(g_\pm^2)| - \frac{1}{2h^2} \int g_\pm^2 \right\}.$$

Add the singular term in $G(f)$ and use (20) and $\lambda \le 1$:

$$\begin{aligned} G(f) &\ge \sum_{\pm} \left\{ \frac{\lambda}{2h} |D(g_{\pm}^2)|(Q_d) - \frac{1}{2h^2} \int g_{\pm}^2 \right\} \\ &\ge \left(\frac{\lambda H(q)}{2h} - \frac{1}{2h^2} \right) \int (g_-^2 + g_+^2). \end{aligned} \tag{21}$$

The second line is (16), applied separately to g_-^2 and g_+^2, whose positivity sets have measure at most q. On $\{f = 0\}$ one has $g_-^2 = h^2$, and on $\{f = 1\}$ one has $g_+^2 = h^2$. Hence $\int (g_-^2 + g_+^2) \ge h^2 s(f)$. Noting that the coefficient in the last line is $C(h,q)/h^2 \ge 0$, this proves $G(f) \ge C(h,q)s(f)$. □

Based on lemmas above, we now establish local optimality of slabs.

Theorem 4 (Local optimality of slabs)**.** *For every compact $K \subset (0,1)$ there exists $\delta_K > 0$, independent of n, such that any finite-perimeter set $E \subset Q_n$ of volume $a \in K$ satisfying $|E \triangle S_{i,a}^{\pm}| \le \delta_K$ has $P_n(E) \ge 1$. Here*

$$S_{i,a}^- = \{x_i < a\} \cap Q_n, \qquad S_{i,a}^+ = \{x_i > 1 - a\} \cap Q_n.$$

Equality occurs only for coordinate slabs. For the explicit choices

$$K = [49/100, 51/100], \qquad \delta_K = 10^{-4}, \tag{22}$$

for $n \ge 2$ one has, in coordinates adapted to the nearby slab,

$$P_n(E) \ge 1 + \frac{1}{401} G(f). \tag{23}$$

Remark 5. *In contrast, in the discrete hypercubes, local optimality of subcubes in maximizing noise stability (a probability-generalized version of the isoperimetric problem) was established by the author [11].*

Proof. In dimension one, every set with volume strictly between zero and one has relative perimeter at least one. Relative perimeter one means that its characteristic function has exactly one interior jump, so it is $(0,a)$ or $(1-a,1)$ up to null sets. Assume henceforth that $n \ge 2$. Coordinate permutations and reflections preserve both relative perimeter and volume; we may therefore use $S_{1,a}^- = \{x_1 < a\}$ and let $f(y) = |E_y|$. Fubini's theorem and the inequality between an integral and the integral of its absolute value give

$$\mathbb{E} f = |E| = a, \qquad |f(y) - a| = \left| \int_0^1 (\mathbf{1}_E(x,y) - \mathbf{1}_{\{x<a\}})\, dx \right| \le |E_y \triangle (0,a)|.$$

Integrating shows $\int |f - a| \le |E \triangle S_{1,a}^-|$.

For a nonempty compact $K \subset (0,1)$, let $m = \min_{a\in K} \min(a, 1-a) > 0$ and set $h = m/2$. Choose $0 < q < 1/2$ such that $hH(q) \geq 10/\sqrt{3}$, which is possible because $H(q) \to \infty$. Set $\delta_K = (m/2)q$. Let $E \subset Q_n$ be any finite-perimeter set of volume $a \in K$ satisfying $|E \triangle S^-_{1,a}| \leq \delta_K$. Then, $\int |f - a| \leq (m/2)q$. Moreover, $\min(a, 1-a) - h \geq m/2$, we have $\int |f - a| \leq (\min(a, 1-a) - h)q$. So the saturation lemma (Lemma 3) applies. Its coefficient satisfies

$$C(h, q) = \frac{\sqrt{3}hH(q)}{4} - \frac{1}{2} \geq 2.$$

Thus $s(f) \leq G(f)/2$ and the fiber-volume bound (Lemma 2) gives $P_n(E) \geq 1+G(f)-s(f) \geq 1 + G(f)/2$. For the empty compact set the assertion is vacuous.

For the explicit interval $[49/100, 51/100]$, set $m = 49/100$, $h = 9/20$ (with better choice than $h = m/2$), and $q = 1/400$. Now $\min(a, 1-a) - h \geq 49/100 - 9/20 = 1/25$ and $\delta_K = (1/25)q = 10^{-4}$. Appendix A proves

$$H(1/400) > \frac{1897}{246}, \qquad C(9/20, 1/400) > \frac{401}{400}. \tag{24}$$

Consequently $s(f) \leq 400G(f)/401$ and $P_n(E) \geq 1 + G(f)/401$, proving (23).

It remains to justify equality for arbitrary BV competitors. In either version, $P_n(E) = 1$ forces $G(f) = 0$. Since $\psi(t) = 0$ only at $t = 0$, both $\nabla f = 0$ almost everywhere and $D^s f = 0$; hence $Df = 0$. A BV function with zero distributional derivative on a connected open cube is constant almost everywhere (for example, mollify on interior cubes and use zero gradient). Its mean fixes the constant as $f = a$. Since $0 < a < 1$, slicing now gives $p(y) \geq 1$ almost everywhere. On the other hand,

$$1 = P_n(E) \geq \int p(y)\, dy = \int_{\partial^* E \cap Q_n} |\nu_x|\, d\mathcal{H}^{n-1} \geq 1.$$

Thus $\int (1 - |\nu_x|)\, d\mathcal{H}^{n-1} = 0$. Because $|\nu_E| = 1$, this implies $\nu_y = 0$ almost everywhere on the reduced boundary, and all transverse distributional derivatives of $\mathbf{1}_E$ vanish. By BV slicing in the transverse variables, for almost every x the function $y \mapsto \mathbf{1}_E(x, y)$ has zero derivative on the connected cube Q_{n-1}. It is therefore a constant equal to either zero or one. Hence $E = A \times Q_{n-1}$ almost everywhere for a measurable $A \subset (0, 1)$. Product variation gives $\mathrm{Per}(A; (0, 1)) = P_n(E) = 1$ and $|A| = a$. The one-dimensional classification above gives $A = (0, a)$ or $(1 - a, 1)$, completing the equality statement. □

Remark 6 (Quantitative fiber control)**.** *Concavity of J, together with $J(0) = J(1) = 0$ and $J(1/2) = 1$, gives $J(t) \geq 2\min(t, 1 - t)$. Thus (4) implies the cube Cheeger bound* $\mathrm{Per}(A; Q_d) \geq 2\min(|A|, 1 - |A|)$. *Let m_f be a median of a real-valued $f \in BV(Q_d)$, so that $|\{f > m_f\}| \leq 1/2$ and $|\{f < m_f\}| \leq 1/2$. Coarea, applied above and below the median, and*

layer-cake give

$$\begin{aligned}|Df|(Q_d) &= \int_0^\infty [\mathrm{Per}(\{f > m_f + t\}; Q_d) + \mathrm{Per}(\{f < m_f - t\}; Q_d)]\, dt \\ &\geq 2\int_0^\infty [|\{f > m_f + t\}| + |\{f < m_f - t\}|]\, dt \\ &= 2\int |f - m_f|.\end{aligned}$$

The coarea identity ignores exceptional threshold values of one-dimensional measure zero. Since $|\mathbb{E}f - m_f| \leq \int |f - m_f|$*, the triangle inequality implies* $\int |f - \mathbb{E}f| \leq 2\int |f - m_f| \leq |Df|(Q_d)$*.*

To control the variation by G*, put* $b = \int |\nabla f|$ *and* $c = |D^s f|(Q_d)$*. Minkowski's integral inequality for* $(1, |\nabla f|)$ *gives*

$$1 + G(f) = \int \sqrt{1 + |\nabla f|^2} + c \geq \sqrt{1 + b^2} + c \geq \sqrt{1 + (b + c)^2}.$$

The last inequality is the triangle inequality for the vectors $(1, b)$ *and* $(0, c)$*. Since* $b + c = |Df|(Q_d)$*, under* (22) *we obtain*

$$P_n(E) - 1 \geq \frac{1}{401}\left(\sqrt{1 + \left(\int |f - a|\right)^2} - 1\right).$$

This controls fiber-volume variation. A bound on the full set distance additionally requires control of fiber arrangement.

4 Quantitative rigidity of slabs

We now establish a quantitative stability theorem for coordinate half-cubes.

Theorem 7 (Quantitative rigidity of slabs)**.** *Let* $E \subset Q_n$ *have finite relative perimeter, with*

$$P_n(E) \leq 1, \qquad \epsilon = \left||E| - \frac{1}{2}\right| \leq 10^{-3}.$$

Then

$$\min_{i,\pm} |E \triangle S^{\pm}_{i,|E|}| \leq 40\sqrt{\epsilon}. \tag{25}$$

The inequality is strict if $\epsilon > 0$*. At* $\epsilon = 0$*,* E *is a coordinate half-cube.*

Proof. Write $a = |E|$, $z = \Phi^{-1}(a)$, $t = |z|$, $F = T^{-1}(E)$, and $p = p_\gamma(F)$. Recall $W(x, \nu) = \sqrt{\sum_i \nu_i^2 e^{x_i^2}}$ defined in (2). Set

$$R = \int_{\partial^* F} (W - 1)\, d\sigma_F, \qquad D = p - \varphi(z).$$

Both D and R are nonnegative, and (3) and $P_n(E) \le 1$ give

$$p + R = \frac{P_n(E)}{\kappa} \le \kappa^{-1}, \qquad D + R \le \kappa^{-1}(1 - e^{-t^2/2}).$$

Using $1 - e^{-s} \le s$ and $\kappa > 5/2$ yields

$$p < \frac{2}{5}, \qquad 0 \le D, R \le \frac{t^2}{5}. \tag{26}$$

The bounds on π and κ used here and below are proved in Appendix A.

Step 1: the threshold and Gaussian deficits. Since $\kappa < 251/100$ and $e^{-s} > 1 - s$ for $s > 0$,

$$\varphi(1/100) > \frac{100}{251}\frac{19999}{20000} = \frac{19999}{50200} > \frac{3}{8}.$$

Thus $\Phi(1/100) - 1/2 > 3/800 > 10^{-3}$. Symmetry and $\epsilon = |a - 1/2| \le 10^{-3}$ imply $t < 1/100$. As $\varphi(s) \ge \varphi(1/100) > 3/8$ on $[0, t]$, $\epsilon = \int_0^t \varphi(s)\, ds \ge (3/8)t$. In particular,

$$t < 1/100, \qquad t \le (8/3)\epsilon \le 3\epsilon. \tag{27}$$

Recall $B(F) = \int_F x\, d\gamma_n$ and $\beta = \varphi(z) - |B(F)|$ defined in (6). By (7) and (26),

$$\beta \le 160\pi^3(1 + z^2)D \le 32(22/7)^3(1 + 10^{-4})t^2 \le 1000t^2. \tag{28}$$

Since $\varphi(z) > 3/8$ and $\beta < 1/10$, $|B(F)| = \varphi(z) - \beta > 11/40 > 0$. Let $\omega = -B(F)/|B(F)|$ and $H = H_{\omega,z}$. Equation (8) gives

$$\eta := \gamma_n(F \triangle H) \le 2e^{-z^2/4}\sqrt{\beta} \le 2\sqrt{\beta} \le 64t. \tag{29}$$

Using $\int_{\partial^* F} \nu_F\, d\sigma_F = -B(F)$ from (10), we obtain the exact normal-excess identity

$$\begin{aligned} e &:= \int_{\partial^* F} |\nu_F - \omega|^2\, d\sigma_F \\ &= 2p - 2\omega \cdot \int_{\partial^* F} \nu_F\, d\sigma_F = 2(p - |B(F)|) = 2(D + \beta) \le 2001t^2. \end{aligned} \tag{30}$$

All these quantities vanish at $t = 0$.

Step 2: a bounded witness on the boundary of F. For a unit vector v define

$$q_v(x) = \sum_i v_i^2(1 - e^{-x_i^2/2}), \qquad 0 \le q_v \le 1.$$

At a boundary point put $\nu = \nu_F(x)$. Cauchy–Schwarz with weights ν_i^2 gives $W \ge \sum_i \nu_i^2 e^{x_i^2/2}$. For $s \ge 0$, $e^s - 1 \ge 1 - e^{-s}$, and thus, $W - 1 \ge q_\nu$. Writing $b_i = 1 - e^{-x_i^2/2} \in [0, 1]$, another Cauchy–Schwarz estimate gives

$$|q_\omega - q_\nu| = \left|\sum_i b_i(\omega_i - \nu_i)(\omega_i + \nu_i)\right| \le |\omega - \nu|\,|\omega + \nu| \le 2|\omega - \nu|.$$

Integrating and using the normal-excess bound gives

$$Q_F := \int_{\partial^* F} q_\omega \, d\sigma_F \le \int_{\partial^* F} (q_{\nu_F} + 2|\omega - \nu_F|) \, d\sigma_F \le R + 2 \int_{\partial^* F} |\omega - \nu_F| \, d\sigma_F \le R + 2\sqrt{pe}. \tag{31}$$

Step 3: transfer to the aligned half-space. Set $X = q_\omega \omega$ and $r_\omega = \operatorname{div}_\gamma X = \omega \cdot \nabla q_\omega - (\omega \cdot x) q_\omega$. Since $\partial_i q_\omega = \omega_i^2 x_i e^{-x_i^2/2}$ and $\sup_s |s| e^{-s^2/2} = e^{-1/2}$,

$$|\omega \cdot \nabla q_\omega| \le e^{-1/2} \sum_i |\omega_i|^3 \le e^{-1/2}.$$

The last inequality uses $|\omega_i| \le 1$ and $\sum_i \omega_i^2 = 1$. In particular $|r_\omega| \le e^{-1/2} + |\omega \cdot x|$ is integrable.

Equal volumes and alignment give the exact identity

$$\begin{aligned} \int_{F \triangle H} |\omega \cdot x - z| \, d\gamma_n &= \int (\mathbf{1}_F - \mathbf{1}_H)(\omega \cdot x - z) \, d\gamma_n \\ &= \omega \cdot B(F) - \omega \cdot B(H) = \varphi(z) - |B(F)| = \beta, \end{aligned} \tag{32}$$

where $B(H) = -\varphi(z)\omega$ and the term involving z cancels because the volumes agree. Therefore

$$\int_{F \triangle H} |\omega \cdot x| \, d\gamma_n \le t\eta + \beta,$$

and

$$\left| \int_F r_\omega \, d\gamma_n - \int_H r_\omega \, d\gamma_n \right| \le (e^{-1/2} + t)\eta + \beta.$$

Formula (9) applies to $X = q_\omega \omega$, whose norm is at most one. On ∂H the outward normal is ω, so

$$\int_H r_\omega \, d\gamma_n = \int_{\partial H} q_\omega \, d\sigma_H =: Q_H.$$

On $\partial^* F$, $q_\omega \ge 0$ and $\omega \cdot \nu_F \le 1$, so

$$\int_F r_\omega \, d\gamma_n = \int_{\partial^* F} q_\omega (\omega \cdot \nu_F) \, d\sigma_F \le \int_{\partial^* F} q_\omega \, d\sigma_F = Q_F.$$

Combining these facts yields

$$Q_H \le \int_F r_\omega \, d\gamma_n + \left| \int_H r_\omega \, d\gamma_n - \int_F r_\omega \, d\gamma_n \right| \le R + 2\sqrt{pe} + (e^{-1/2} + t)\eta + \beta. \tag{33}$$

For $t > 0$, the preceding estimates and $e^{-1/2} < 2/3$ give

$$\frac{Q_H}{t} < \frac{1}{500} + 57 + 64 \left(\frac{2}{3} + \frac{1}{100} \right) + 10 < 111.$$

At $t = 0$ all terms in (33) vanish. Hence $Q_H \le 111t$ in all cases.

Step 4: concentration of the normal on a coordinate. Let X_0 be standard Gaussian in $\mathbb{R}^n$. Its decomposition $X_0 = (\omega \cdot X_0)\omega + Z$ has independent components, where Z is centered (degenerate) Gaussian with covariance $\mathrm{Id} - \omega \otimes \omega$. Thus, conditionally on $\omega \cdot X_0 = z$, the ith coordinate of X_0 has mean $\omega_i z$ and variance $1 - \omega_i^2$. For a Gaussian Y with mean m and variance s^2, completion of the square gives

$$\mathbb{E}e^{-Y^2/2} = \frac{1}{\sqrt{1+s^2}} \exp\left(-\frac{m^2}{2(1+s^2)}\right);$$

the same formula holds for $s = 0$ by direct evaluation. Consequently,

$$\mathbb{E}[1 - e^{-X_{0,i}^2/2} \mid \omega \cdot X_0 = z] = 1 - \frac{\exp\{-\omega_i^2 z^2/[2(2-\omega_i^2)]\}}{\sqrt{2-\omega_i^2}} \geq (1 - 2^{-1/2})(1 - \omega_i^2).$$

For the inequality, discard the exponential factor and use the concavity of $b(v) = 1 - (1+v)^{-1/2}$ on $[0,1]$: $b''(v) = -3(1+v)^{-5/2}/4 < 0$, so $b(v) \geq vb(1) = v(1 - 2^{-1/2})$.

Gaussian integration on the plane factors as $\varphi(z)$ times the conditional Gaussian law. It follows that

$$\begin{aligned} Q_H &= \int_{\partial H} q_\omega \, d\sigma_H = \varphi(z)\mathbb{E}[q_\omega(X_0) \mid \omega \cdot X_0 = z] \geq \varphi(z)(1 - 2^{-1/2}) \sum_i \omega_i^2(1 - \omega_i^2) \\ &= \varphi(z)(1 - 2^{-1/2}) \left(1 - \sum_i \omega_i^4\right) \geq \frac{1}{10}\left(1 - \max_i \omega_i^2\right). \end{aligned} \tag{34}$$

Indeed $\sum_i \omega_i^4 \leq (\max_i \omega_i^2) \sum_i \omega_i^2$, $\varphi(z) > 3/8$, and $1 - 2^{-1/2} > 4/15$. Put $u = 1 - \max_i \omega_i^2$. We have $u \leq 10 Q_H \leq 1110t$. If $t = 0$, then $u = 0$ and $\eta = 0$; therefore ω is a signed coordinate vector e_i or $-e_i$ for some i, and $F = H$ almost everywhere. This proves the half-volume equality statement.

Step 5: comparison with a same-volume coordinate half-space. Choose an index attaining $\max_i |\omega_i|$ and let ξ be the corresponding signed coordinate unit vector with $\omega \cdot \xi = \max_i |\omega_i| = \sqrt{1-u}$. Their angle is $\theta = \arcsin\sqrt{u} \in [0, \pi/2]$. If ω and ξ are not collinear, the orthogonal projection of a standard Gaussian vector onto $\mathrm{span}\{\omega, \xi\}$ has the standard two-dimensional Gaussian distribution, which is rotationally invariant. Its polar angle is uniform and independent of its radius. The central half-spaces disagree on two sectors of angle θ, hence $\gamma_n(H_{\omega,0} \triangle H_{\xi,0}) = \theta/\pi$. For collinear vectors, both sides are zero. Also, for any unit v, the nested half-spaces at thresholds 0 and z have symmetric-difference measure $|\Phi(z) - 1/2| = \epsilon$. The triangle inequality therefore gives

$$\begin{aligned} \gamma_n(H_{\omega,z} \triangle H_{\xi,z}) &\leq \gamma_n(H_{\omega,z} \triangle H_{\omega,0}) + \gamma_n(H_{\omega,0} \triangle H_{\xi,0}) + \gamma_n(H_{\xi,0} \triangle H_{\xi,z}) \\ &\leq \theta/\pi + 2\epsilon \leq \tfrac{1}{2}\sqrt{u} + t, \end{aligned}$$

where $\arcsin r \le (\pi/2)r$ on $[0,1]$ and $2\epsilon = 2\int_0^t \varphi(s)\,ds \le 2\varphi(0)t < t$. Thus, for $\epsilon > 0$,

$$\begin{aligned}\gamma_n(F\triangle H_{\xi,z}) &\le \eta + \gamma_n(H_{\omega,z}\triangle H_{\xi,z})\\ &\le 65t + \tfrac{1}{2}\sqrt{1110t}\\ &\le 195\epsilon + \tfrac{1}{2}\sqrt{3330\epsilon} < 40\sqrt{\epsilon},\end{aligned}$$

where the last inequality follows since $\epsilon \le 1/1000$. Finally, if $\xi = e_i$, $T(H_{\xi,z}) = S^-_{i,a}$; if $\xi = -e_i$, $T(H_{\xi,z}) = S^+_{i,a}$, since $\Phi(-z) = 1-a$. The map T preserves the measure of symmetric differences. This proves (25). □

The proof also gives qualitative rigidity for sequences with $|E_k| \to 1/2$ and $P_{n_k}(E_k) \to 1$, even when some perimeters exceed one. To make the assertion precise, set $a_k = |E_k|$, $z_k = \Phi^{-1}(a_k) \to 0$, and

$$\tau_k = \frac{P_{n_k}(E_k)}{\kappa} - \varphi(z_k) \ge 0.$$

With the notation of the proof, the area formula gives $D_k + R_k = \tau_k \to 0$. Hence $D_k, R_k \to 0$, and (7) implies $\beta_k \to 0$. Eventually $|B(F_k)| = \varphi(z_k) - \beta_k > 0$, so ω_k is defined. Then (8) gives $\eta_k \to 0$, (30) gives $e_k \to 0$, and p_k stays bounded. The right side of (33) tends to zero. Since $\varphi(z_k)(1-2^{-1/2})$ is bounded below by a positive constant, the first inequality in (34) forces $1 - \max_i \omega_{k,i}^2 \to 0$. The final half-space comparison, now with error $\eta_k + \frac{1}{2}\sqrt{1-\max_i \omega_{k,i}^2} + 2|a_k - 1/2|$, tends to zero. After applying T, this says that the distance to some same-volume coordinate slab tends to zero, uniformly over the varying dimensions.

5 An explicit central plateau

Theorem 8 (Central plateau with explicit width)**.** *Set*

$$\varepsilon_0 = \frac{1}{160\,000\,000\,000} = 6.25 \times 10^{-12}.$$

For every $n \ge 1$ and every a with $|a - 1/2| \le \varepsilon_0$,

$$I_n(a) = 1. \tag{35}$$

A finite-perimeter set $E \subset Q_n$ of volume a has $P_n(E) = 1$ if and only if it is a coordinate slab, up to null sets. Also $I_\infty(a) := \inf_{n\ge 1} I_n(a) = 1$ on this interval.

Proof. Coordinate slabs have volume a and relative perimeter one, so $I_n(a) \le 1$. Consider any admissible competitor E. If $P_n(E) > 1$ (including infinite perimeter), there is nothing

to prove. If $P_n(E) \le 1$, Theorem 7 applies because $|a - 1/2| \le \varepsilon_0 < 10^{-3}$ and supplies a same-volume slab S such that

$$|E \triangle S| \le 40\sqrt{\varepsilon_0} = 10^{-4}.$$

The volume lies in $[49/100, 51/100]$, so Theorem 4 gives $P_n(E) \ge 1$. Every competitor therefore has perimeter at least one, and thus, $I_n(a) = 1$. If $P_n(E) = 1$, the equality statement in Theorem 4 forces E to be a coordinate slab. Conversely, every such slab has perimeter one. Taking the infimum over n gives $I_\infty(a) = 1$ on the same interval. □

Table 1 summarizes the dimension-independent constants. The small numerical width reflects a conservative combination of the published Gaussian constant and a square-root rigidity modulus. It is an admissible value of ε, not the maximal plateau half-width.

Estimate	Constant	Role
Small-support bound	$H(1/400) > 1897/246$	Saturation cost
Slab neighborhood	$\delta_K = 10^{-4}$	Local minimality
Local gain	$G(f)/401$	Equality rigidity
Slab distance	$40\sqrt{\lvert a - 1/2\rvert}$	Global rigidity for $P \le 1$
Plateau half-width	$\varepsilon_0 = \delta_K^2/1600$	Theorem 8

Table 1: Constants proving the central plateau.

6 Conjectures for the limiting profile

We now record two conjectures that extend the central plateau theorem. Throughout this section,

$$I_\infty(a) = \inf_{n \ge 1} I_n(a), \qquad v = \min(a, 1 - a).$$

Let $\omega_k = \pi^{k/2}/\Gamma(1 + k/2)$ be the volume of the Euclidean unit ball in $\mathbb{R}^k$, and define

$$c_k = \frac{k}{2}\omega_k^{1/k}, \qquad B_k(v) = c_k v^{1-1/k} \quad (v > 0). \tag{36}$$

In particular, $B_1(v) = 1$, $B_2(v) = \sqrt{\pi v}$, and $B_3(v) = 3(\pi/6)^{1/3}v^{2/3}$. The remaining coefficients are

$$c_4 = 2(\pi^2/2)^{1/4}, \qquad c_5 = \frac{5}{2}(8\pi^2/15)^{1/5}, \qquad c_6 = 3(\pi^3/6)^{1/6}.$$

The geometric interpretation is a positive-orthant ball in k coordinates, extended cylindrically in the remaining coordinates. Its volume and relative perimeter are

$$v = \frac{\omega_k r^k}{2^k}, \qquad P = \frac{k\omega_k r^{k-1}}{2^k} = B_k(v).$$

This interpretation requires $r \le 1$, equivalently $v \le \omega_k/2^k$, so that the ball does not cross the opposite cube faces. The factors 2^{-k} arise because the orthant occupies 2^{-k} of a ball, and the coordinate-plane boundaries are excluded from relative perimeter. A product with unused coordinates preserves both this volume and perimeter.

Conjecture 1 (Seven-branch limiting profile)**.** *For every $a \in (0,1)$,*

$$I_\infty(a) = \min\{\pi I_{\mathrm{G}}(a), B_1(v), B_2(v), B_3(v), B_4(v), B_5(v), B_6(v)\}, \qquad v = \min(a, 1-a). \tag{37}$$

At the endpoints, $I_\infty(0) = I_\infty(1) = 0$.

The conjecture concerns the value of the infinite-dimensional infimum. It does not assert that the infimum is attained in a fixed finite dimension on the Gaussian branch, nor that every finite-dimensional minimizer is a cylindrical orthant ball. Such a finite-dimensional classification is a different statement; indeed, Milman [9, Theorem 1.6] disproves the general tubular-face classification for cubes of dimension at least ten.

Solving $B_k(\tau_k) = B_{k-1}(\tau_k)$ gives

$$\tau_k = \left(\frac{k-1}{k}\right)^{k(k-1)} \frac{\omega_{k-1}^k}{\omega_k^{k-1}}.$$

In particular,

$$\tau_6 = \frac{5^{24}}{2^7 3^{31}\pi^3} \simeq 0.024314291553, \quad \tau_5 = \frac{2^{23}3^4\pi^2}{5^{16}} \simeq 0.043949566442,$$
$$\tau_4 = \frac{6561}{8192\pi^2} \simeq 0.081148472397, \quad \tau_3 = \frac{4\pi}{81} \simeq 0.155140377955, \qquad \tau_2 = \frac{1}{\pi} \simeq 0.318309886184.$$

The Gaussian-to-six-dimensional crossing is denoted by τ_{G} and satisfies

$$\pi I_{\mathrm{G}}(\tau_{\mathrm{G}}) = B_6(\tau_{\mathrm{G}}), \qquad \tau_{\mathrm{G}} \simeq 0.015890647092. \tag{38}$$

Theorem 9 (Upper bound and small-volume asymptotics [10, 12])**.** *For relative Euclidean perimeter on unit cubes,*

$$I_\infty(a) \le \pi I_{\mathrm{G}}(a) \qquad a \in [0,1],$$

and

$$I_\infty(a) \sim \pi I_{\mathrm{G}}(a) \sim \pi a\sqrt{2\log(1/a)} \qquad (a \downarrow 0). \tag{39}$$

Moreover, the upper bound $\pi I_{\mathrm{G}}(a)$ is asymptotically attained by the set

$$\left\{ x \in Q_n : \sqrt{\frac{2}{n}} \sum_{i=1}^{n} \cos(\pi x_i) \le \Phi^{-1}(a) \right\}. \tag{40}$$

Volume range	Proposed value	Branch
$0 < v \le \tau_{\mathrm{G}}$	$\pi I_{\mathrm{G}}(v)$	Gaussian
$\tau_{\mathrm{G}} \le v \le \tau_6$	$B_6(v)$	Six-dimensional orthant ball
$\tau_6 \le v \le \tau_5$	$B_5(v)$	Five-dimensional orthant ball
$\tau_5 \le v \le \tau_4$	$B_4(v)$	Four-dimensional orthant ball
$\tau_4 \le v \le \tau_3$	$B_3(v)$	Three-dimensional orthant ball
$\tau_3 \le v \le 1/\pi$	$\sqrt{\pi v}$	Quarter-disk cylinder
$1/\pi \le v \le 1/2$	1	Coordinate slab

Table 2: Predicted branches of Conjecture 1. Reflection about $a = 1/2$ gives the other half of the profile.

This theorem is the cube specialization of [10, Theorem 5] and [12, Theorems 1.2 and 1.5, Example 4]. Theorem 9 establishes the leading small-volume behavior predicted by Conjecture 1. The exact identity $I_\infty(a) = \pi I_{\mathrm{G}}(a)$ for sufficiently small positive a, predicted by the conjecture, is stronger than (39).

For $0 < a < \tau_{\mathrm{G}}$, the upper bound $\pi I_{\mathrm{G}}(a)$ is strictly smaller than $B_k(a)$ for every $k \ge 1$. Consequently, there exist sets in sufficiently high-dimensional cubes whose perimeters are strictly smaller than those of cylindrical orthant balls. This provides an explicit confirmation of the phenomenon established in [9, Theorem 1.6], namely, that cylindrical orthant balls need not be isoperimetric minimizers.

A Analytic certificates for the constants

We verify (24) by rational inequalities, without numerical inversion of the Gaussian distribution.

A.1 Elementary bounds

We first prove $25/8 < \pi < 22/7$. Polynomial division gives

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}.$$

Integrating on $[0,1]$, using $4\int_0^1 (1+x^2)^{-1}\,dx = \pi$, yields

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{22}{7} - \pi.$$

For the lower bound, the tangent addition formula and the fact that both angles lie in $(0, \pi/2)$ give $\arctan 1 = \arctan(1/2) + \arctan(1/3)$. The alternating expansion $\arctan x =$

$x - x^3/3 + x^5/5 - \cdots$ has a positive remainder after an even number of terms. Thus

$$\pi > 4\left(\frac{1}{2} - \frac{1}{24} + \frac{1}{160} - \frac{1}{896} + \frac{1}{3} - \frac{1}{81}\right) = \frac{284663}{90720} = \frac{25}{8} + \frac{1163}{90720} > \frac{25}{8}.$$

Since $\kappa^2 = 2\pi$, these bounds imply $\kappa > 5/2$; also

$$(251/100)^2 - \frac{44}{7} = \frac{1007}{70000} > 0,$$

so $\kappa < 251/100$.

For $x = 98/25$, let $a_j = x^j/j!$. For every $j \geq 9$, $a_{j+1}/a_j = x/(j+1) \leq x/10 < 1$. The tail from a_9 onward is therefore bounded by a geometric series:

$$e^x \leq \sum_{j=0}^{8} \frac{x^j}{j!} + \frac{x^9}{9!}\frac{1}{1 - x/10} = \frac{59212656221511566}{1174163818359375} < \frac{101}{2}. \tag{41}$$

The difference between $101/2$ and the displayed fraction is positive:

$$\frac{165233211273743}{2348327636718750} > 0.$$

Finally, $e^{1/2} > 1 + 1/2 = 3/2$ gives $e^{-1/2} < 2/3$.

A.2 Mills bounds and the small-support estimate

Let $\overline{\Phi} = 1 - \Phi$ and $M(t) = \overline{\Phi}(t)/\varphi(t)$, $t > 0$. Differentiation gives $M' = tM - 1$. An equivalent integral representation is $M(t) = \int_0^\infty e^{-tu-u^2/2}\,du \leq 1/t$. Define

$$L(t) = \frac{t(t^2+5)}{t^4 + 6t^2 + 3}, \qquad U(t) = \frac{t^2+2}{t^3 + 3t}.$$

Differentiating these rational functions and collecting terms gives

$$L' - tL + 1 = \frac{24}{(t^4 + 6t^2 + 3)^2} > 0, \qquad U' - tU + 1 = -\frac{6}{t^2(t^2+3)^2} < 0.$$

For either $A = L$ or $A = U$, write $r_A = A' - tA + 1$ and $h_A = M - A$. Then $h_A' - th_A = -r_A$, and multiplication by $e^{-t^2/2}$ followed by integration yields

$$M(t) - A(t) = e^{t^2/2}\int_t^\infty r_A(s)e^{-s^2/2}\,ds.$$

The boundary term at infinity vanishes because M, L, and U are $O(1/t)$ there. The signs of the residuals prove

$$\frac{t(t^2+5)}{t^4 + 6t^2 + 3} < M(t) < \frac{t^2+2}{t^3 + 3t}. \tag{42}$$

Both rational bounds are asymptotic to $1/t$, which also establishes the Mills asymptotic used in Section 6.

At $t = 14/5$, direct substitution gives $L(t) = 22470/69691$. Using $\kappa < 251/100$ and (41),

$$\begin{aligned}\overline{\Phi}(14/5) &= \kappa^{-1} e^{-98/25} M(14/5) \\ &> \frac{100}{251}\frac{2}{101}\frac{22470}{69691} = \frac{4494000}{1766736541} > \frac{1}{400}.\end{aligned}$$

For the last comparison the positive difference is $30863459/706694616400$. Since the Gaussian tail is strictly decreasing, $t_q = \Phi^{-1}(1-q) > 14/5$ for $q = 1/400$. By symmetry, $J(q) = \kappa\varphi(t_q)$. The upper Mills bound consequently gives

$$H(q) = \frac{\kappa}{M(t_q)} > \kappa\frac{t_q(t_q^2+3)}{t_q^2+2} > \frac{5}{2}\frac{1897}{615} = \frac{1897}{246}.$$

To justify replacing t_q by $14/5$, note that $k(t) = t(t^2+3)/(t^2+2) = t + t/(t^2+2)$ and $k'(t) = 1 + (2-t^2)/(t^2+2)^2 > 0$: the numerator after multiplication by $(t^2+2)^2$ is $t^4 + 3t^2 + 6 > 0$. Also $k(14/5) = 1897/615$. For $h = 9/20$ we obtain

$$C(h,q) > \frac{17073\sqrt{3}}{19680} - \frac{1}{2} > \frac{401}{400}.$$

The last inequality is equivalent to $\sqrt{3} > 49282/28455$. Both sides are positive, and $3(28455)^2 - (49282)^2 = 345551 > 0$, proving the comparison. This completes the certificate (24).

A.3 Radial-shell checks

Let $a_0 = 81/256$. For $x \in [25/8, 22/7]$,

$$\frac{d}{dx}\frac{x^2}{2x - a_0} = \frac{2x(x-a_0)}{(2x-a_0)^2} > 0.$$

Therefore the upper bound on π gives the lower bound

$$c_* > \frac{12}{25} - \frac{9}{224} - \frac{16(22/7)^2}{63(2(22/7) - 81/256)} = \frac{73871593}{3773901600}.$$

Subtracting $1/100$ gives $36132577/3773901600 > 0$, so $c_* > 1/100$. Both summands of $C = 2/(2\pi - a_0) + 81/(256\pi^2)$ decrease with π. Hence the lower bound on π yields

$$\frac{64C}{63} < \frac{64}{63}\left(\frac{2}{2(25/8) - 81/256} + \frac{81}{256(25/8)^2}\right) = \frac{22448624}{59810625} < \frac{2}{5}.$$

Here $2/5 - 22448624/59810625 = 1475626/59810625 > 0$. These exact rational comparisons prove both shell constants.

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